

Code generation and model creation in PapyrusWeb

This document presents a step-by-step tutorial on how to create UML models/diagrams and how to generate code from these models.

How-to objective

create an UML model with a set of classes and to generate Java code from it.

This how-to is separated into 3 major steps:

1. **Model creation:** how to create UML models and diagrams.
2. **Model edition:** how to edit the model elements.
3. **Code generation:** how to generate code from these models.

The URL of PapyrusWeb is : <https://papyrus-web.deeplab.fr/>

Model Creation

PapyrusWeb enables the creation of different kinds of projects. The start page provides a set of templates, or an existing project may be reopened.

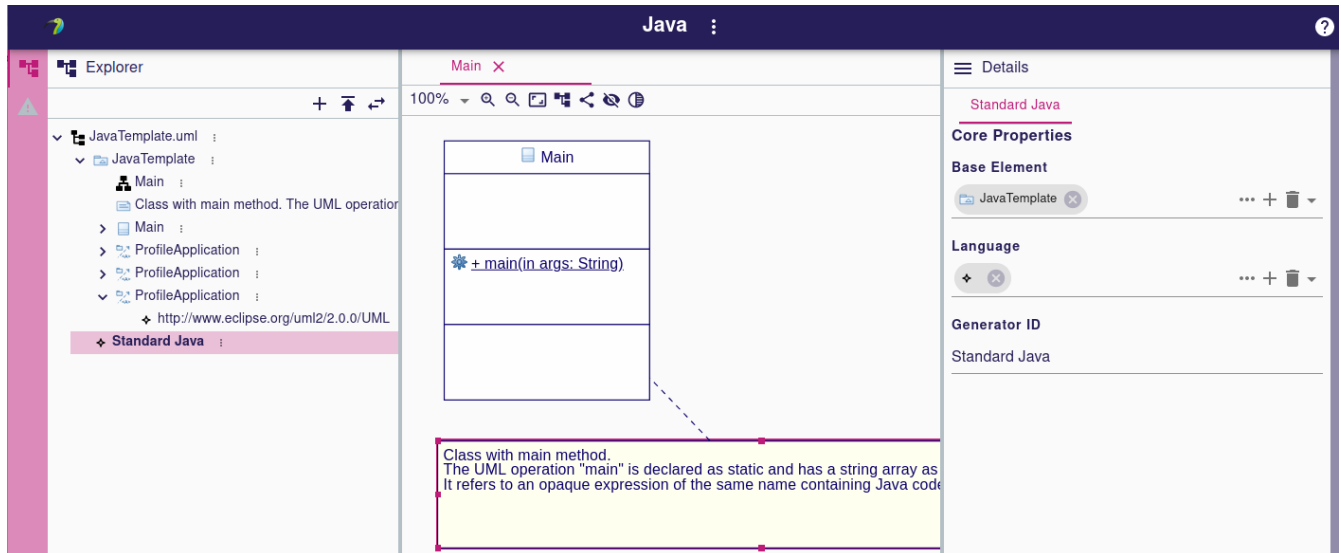
The figure below show the current available templates:



We select the **Java template**. It performs the following actions:

- creates a Java project template with one Class with a main method and a Comment.
- applies 3 profiles necessary for code generation: *standard*, *Codegen* and *PapyrusJava*
- applies the stereotype *GeneratorHint*, setting the generator ID to *Standard Java*.

The project should have a structure as illustrated below:



A project has three main components:

- **Model explorer** (right side): the model explorer enables the navigation and edition in the existing model elements. Each model element has a contextual menu with a specific set of actions. For instance, it enables creating child elements of a given Class.
- **Diagram editor** (center): the diagram editor enables creating visual representation of the elements of the model
- **Property view** (left side): the property view enables the edition of properties of each selected element of the model.

All the possible options are explained in the [PapyrusWeb help documentation](#)

Model edition

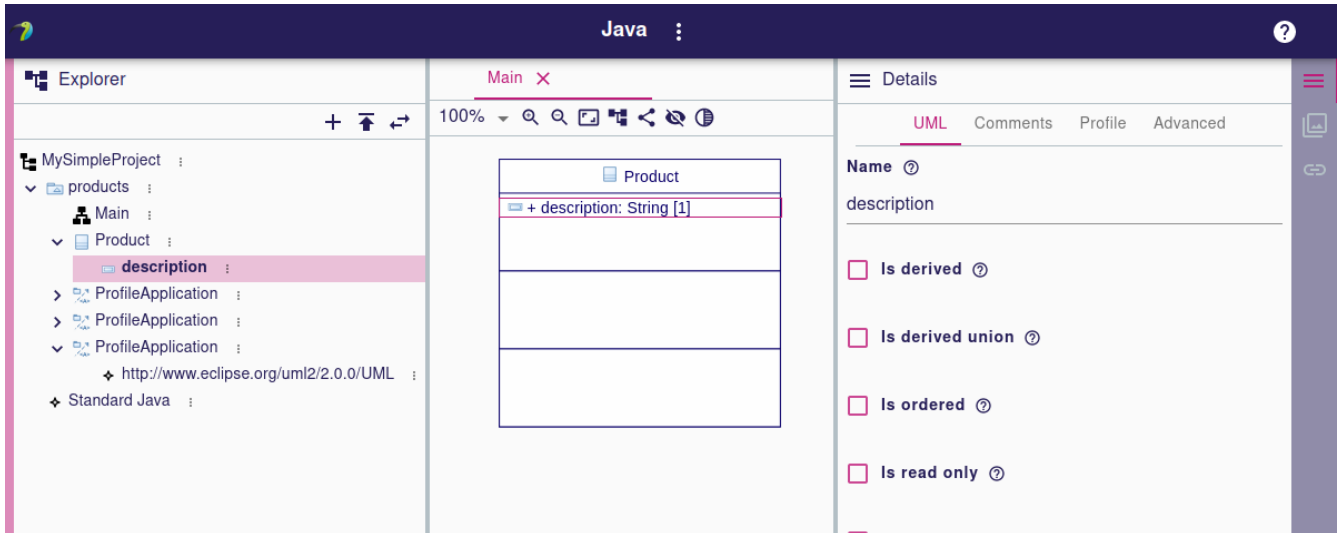
The model may be edited using the diagram editor or the model explorer. Both views are synchronized.

We propose to perform the following actions:

- selection of a comment in the diagram editor and deletion;
- selection on the *Main* class and edition of its name using the property view. Set the name to *Person*.
- selection of the compartment below the class name and inclusion of a property. There is a contextual menu that enables creating a property.

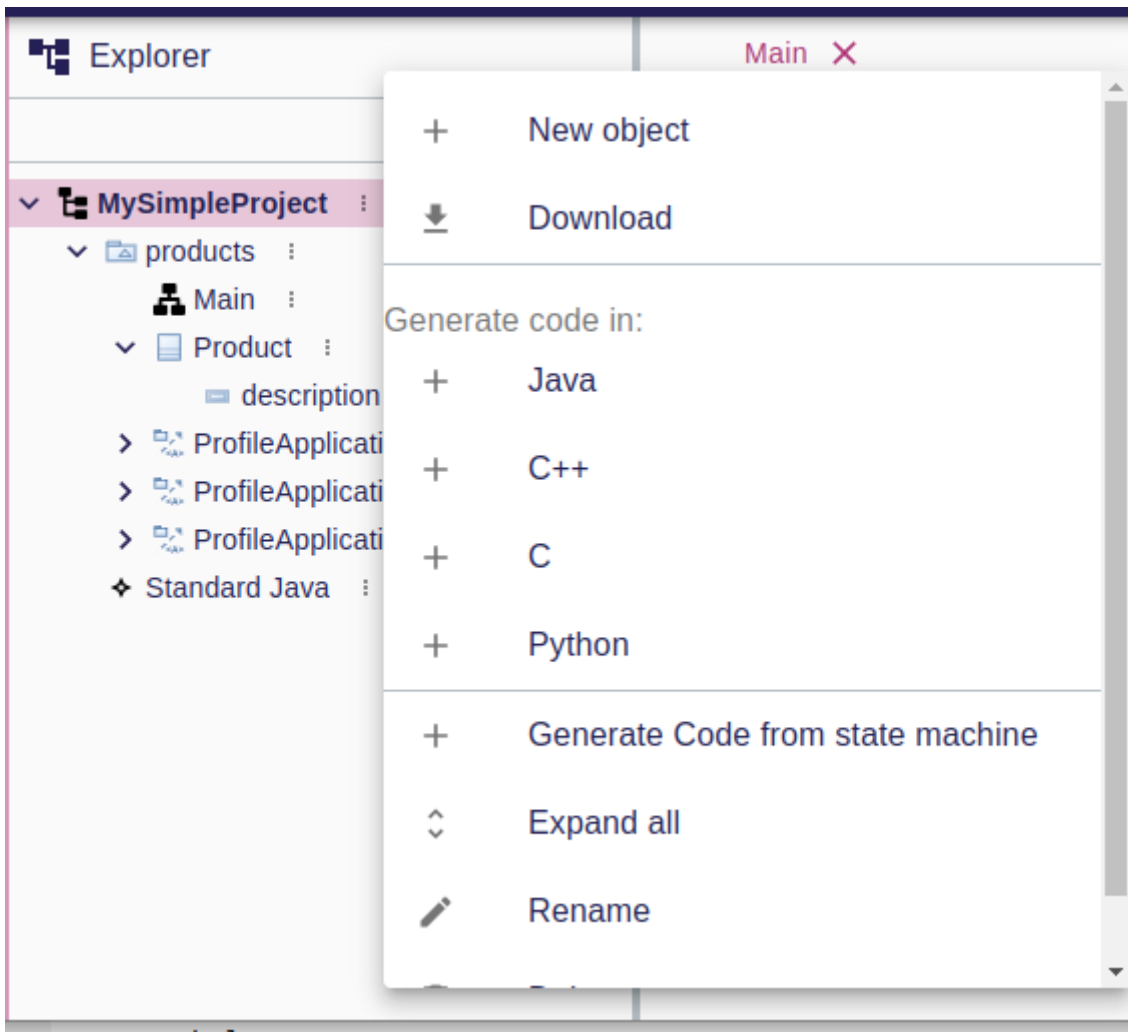
- edition of the property name to **name** and property type to **String**
- deletion of the main method.
- modification of the name of the root element *Model* into *products*, using the property view. The generated project and root package will have this value.
- modification of the root element into *MySimpleProject*. The generated zip file will have this value.

The current model should have the structure illustrated below. 7



Code generation

The last step is to generate the Java code for this simple model. The root element *MySimpleProject* has a contextual menu with the possible code generation actions.



We select **Generate code in Java**. It produces a file *MySimpleProject.zip*, which is an Eclipse project containing the generated code.

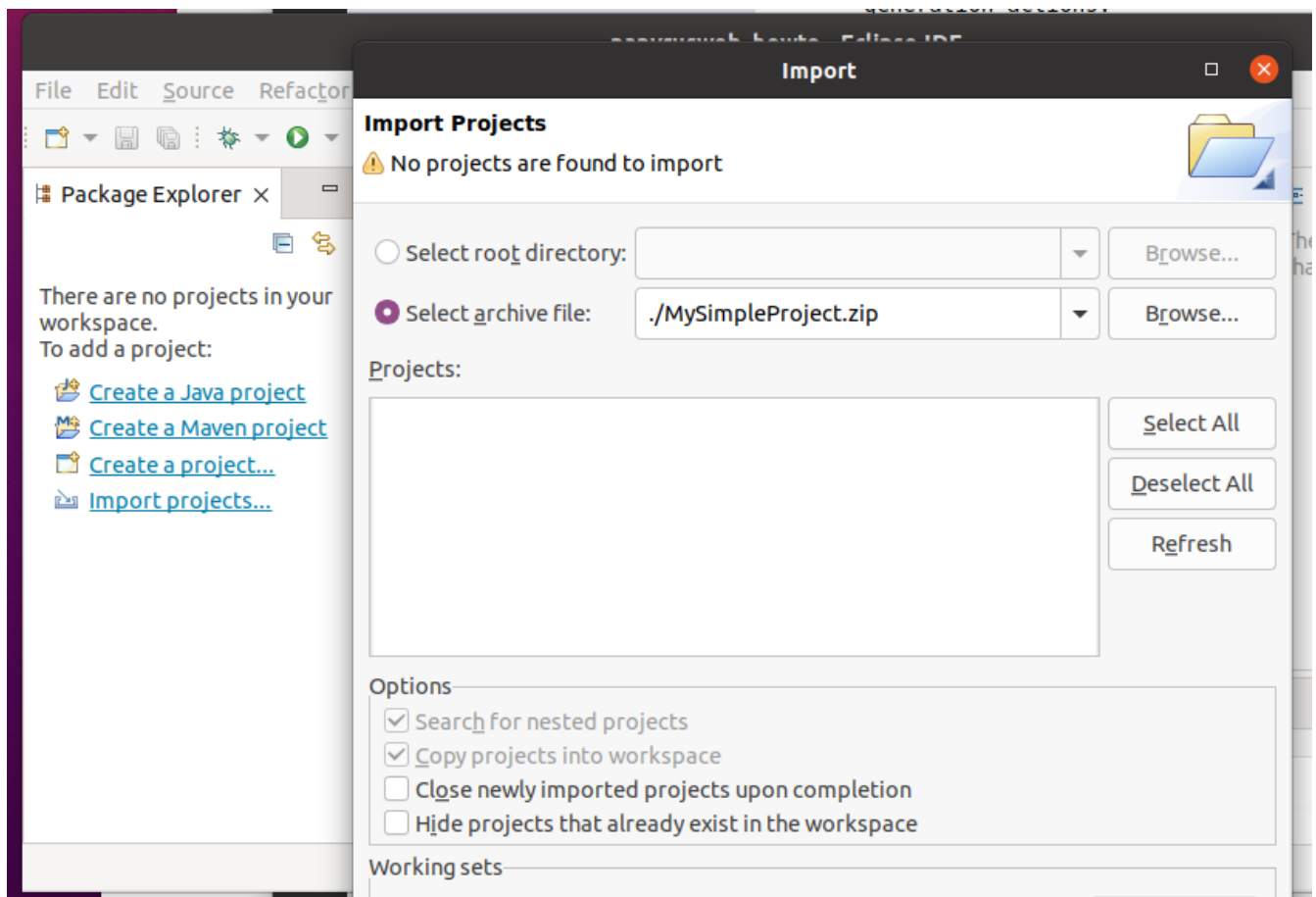
NOTE: The other target languages might also be chosen, but since the project is not configured for these languages, there might errors in the generation process.

Configuring the code generation

The code generation task may be configured by using specific *Code generation profiles*. We do not detailed all the possible generation configurations. They are described in ../description_software_designer/sw_designer.html

Using the generated code

The generated code may be edited in the Eclipse IDE. It is necessary to import the generated zip file using the "Import Projects" menu, as shown below.



Once, the project is imported, it is necessary to add the *Java nature* into the project settings. This is done by updating the **.project** file. The excerpt below can be copied and pasted into the file content, and replacing the *buildSpec* and *nature* tags.

NOTE: The generated code targets the Eclipse platform, which means it generates projects compatible with the Eclipse environment. However, the generated code may be opened in different IDE's, such as VSCode, but the target project configurations need to be